Exercise 1:

- **Data Generating Process:** $Y = 3 + 2X - 0.5X^2 + 0.8X^3 + \varepsilon$, where $n = 100$ and $X, \varepsilon \sim N(0, 1)$.
- **Forward Stepwise Selection (C_p):** Selected features: ('X1', 'X10', 'X2', 'X3').
- **Lasso Regression (5-Fold CV):**
  - Optimal Penalty (α): 0.1261
  - Non-zero Coefficients:
    - $\hat{\beta}_1 = 1.7206$ (True $\beta_1 = 2.0$)
    - $\hat{\beta}_2 = -0.5601$ (True $\beta_2 = -0.5$)
    - $\hat{\beta}_3 = 2.1072$ (True $\beta_3 = 0.8$)

Lasso regression demonstrates exceptional feature selection capabilities in the presence of noise predictors. By applying an penalty, Lasso effectively shrinks all non-informative higher-order polynomial coefficients (X^4 through X^{10}) exactly to zero while accurately recovering the true cubic structure of the underlying process. Forward Stepwise selection correctly identifies the three primary terms (X, X^2, X^3), but retains one spurious high-degree noise term (X^{10}) due to sample variance.

Exercise 2: Predicting College Applications (College Dataset)

1. Performance Summary Table

Method	Test MSE	Test R2 Score	Non-Zero Features
OLS (Ordinary Least Squares)	$1,659,682$	\$90.87\%\$	$17/17$
Ridge Regression	$1,707,146$	\$90.61\%\$	$17/17$
Lasso Regression	$1,753,283$	\$90.36\%\$	$14/17$

PCR (Principal Components Regression)	$1,659,682$	90.87%	17 components
PLS (Partial Least Squares)	$1,671,280$	90.81%	17 components
Baseline Null Model Test MSE: $18,179,067$			

2. Interpretation

All regularized and dimension-reduction techniques substantially outperform the baseline Null model, capturing over 90.3% of the variance in application volume. OLS and PCR achieve the lowest Test MSE on this split. While Lasso yields a marginally higher MSE than OLS, it achieves superior model interpretability by eliminating 3 redundant predictors and producing a sparse model with only 14 active variables.

Exercise 3: Training vs. Test Error as Model Size Grows

1. Experimental Results

- **Optimal Model Size:** 13 predictors
- **Minimum Test MSE:** 1.0823

2. Interpretation

While the Training MSE decreases monotonically as additional predictors are added (due to the model fitting sample-specific noise), the Test MSE displays a classic U-shaped curve. Test MSE reaches its global minimum at exactly 13 predictors. This precisely matches the ground-truth parameter vector β, which contained 13 non-zero entries and 7 zero entries out of 20 total features. Incorporating more than 13 predictors leads to overfitting and degraded generalization.

Exercise 4: Per-Capita Crime Rate Prediction (Boston Dataset)

1. Performance Comparison

- **Ridge CV MSE:** 42.840
- **Lasso CV MSE:** 42.786
- **PCR CV MSE:** 43.290

2. Interpretation & Final Model Justification

Lasso achieves the lowest Cross-Validated MSE (42.786). Therefore, **Lasso is recommended as the final model**. In addition to superior predictive performance, Lasso performs feature selection by zeroing out irrelevant neighborhood statistics, producing a parsimonious model that

is far easier to interpret for urban policy decisions than dense coefficient models like Ridge or PCR.

Exercise 5: Ridge and Lasso Coefficient Paths (Diabetes Dataset)

1. Observations

- **Lasso Path Analysis:** As the regularization penalty parameter λ decreases (moving from left to right on the $-\log\lambda$ scale), key physiological features enter the active model sequentially.
- **Entry Order:** Predictors with strong linear associations with disease progression—such as bmi (body mass index) and s5 (serum triglycerides)—cross the zero boundary first.

2. Interpretation

The Lasso path visually confirms the theoretical property of sparsity: strong predictors enter the model early at high penalty values, while weak predictors remain zeroed out until λ becomes small. This entry order aligns with univariate exploratory data analyses from prior domain studies.

Exercise 6: Impact of Feature Standardisation

1. Coefficient Comparison Table

Feature Name	Standardised β^	Unstandardised β^
AtBat	14.9973	-2.0537
Hits	19.9733	7.0558
HmRun	11.2362	1.0724
Runs	17.8021	-1.3161
RBI	17.6466	0.2706

Walks	19.9261	5.7221
Years	12.9479	−0.6802
CAtBat	19.8541	−0.2086
CHits	22.0901	0.2419
CHmRun	20.6812	0.0296
CRuns	22.6253	1.4657
CRBI	22.8768	0.7052
CWalks	17.3511	−0.7297
League[N]	2.6409	2.3088
Division[W]	−14.3733	−6.2482
PutOuts	18.7466	0.2940
Assists	1.3210	0.3754

Errors	-1.5243	-2.5570
NewLeague[N]	2.1878	1.6295

Ridge regression adds an ℓ_2 penalty based on the sum of squared coefficients ($\lambda \sum \beta_j^2$).

Without standardization, variables measured on large scales (such as career statistics CAtBat and CRuns, which reach thousands) are forced to take tiny unstandardized coefficients to avoid heavy penalization. Consequently, unstandardized Ridge severely distorts feature importance based purely on measurement scale. Standardizing variables to mean zero and unit variance ensures that the shrinkage penalty is applied equitably based on true predictive power.

1. **Regularization suppresses variance:** Ridge and Lasso effectively mitigate multicollinearity and prevent overfitting compared to unregularized OLS in noisy datasets.
2. **Lasso yields sparse models:** Where interpretability is crucial, Lasso provides high accuracy while removing irrelevant features.
3. **Standardization is mandatory:** Shrinkage models are scale-dependent; standardizing features via pipelines before cross-validation prevents data leakage and scale distortion.